

Harrison A. DiAmbrosio

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EDUCATION

Worcester Polytechnic Institute | Worcester, MA

August 2024 - Present

Combined BS/MS. in Computer Science (expected May 2028) | GPA: 3.88

PROJECTS

BeepBoop.nvim | github.com/EggbertFluffle/beepboop.nvim

Technologies: Lua, Zig, C++, Neovim

Neovim plugin designed to provide programmable audio cues for editor events

- Implemented a user-friendly and configuration scheme while staying compliant with popular plugin standards
- Crafted a companion binary leveraged via IPC to bring platform agnostic audio playback to all major operating systems
- Designed a remote audio themes standard based upon Git repositories
- Managed open source community suggestions and contributions while garnering attention via social media

Mezzaluna | github.com/MezzalunaWM/Mezzaluna

Technologies: Lua, Zig, Wayland

A Wayland compositor built in an ongoing partnership with a fellow developer, abstracting compositor complexities behind an approachable, Neovim-inspired configuration API

- Designing an intuitive and expressive Lua API with type completion and robust debug tooling
- Engineering atomic state changes for clients to guarantee frame-perfect resizing and repositioning
- Contributing thorough documentation on configuration API and the development process through technical blogs (eggbert.xyz/blogs/the_perfect_frame)

Conductor | github.com/EggbertFluffle/Conductor

Technologies: Haskell, C, Lua

A domain specific language for describing deterministic windowing patterns, built in a three-person team for a graduate programming languages course

- Co-authored a formal language specification and BNF grammar, defining behaviors for language primitives, runtime parameter evaluation and recursive layout rules
- Authored the JSON-based IPC protocol connecting the Conductor executable to consuming window managers
- Built implementation references for DWL and Mezzaluna, demonstrating the platform-agnostic design

EXPERIENCE

Co-founder and Developer - Lunabyte Studios | Worcester, MA

May 2026 - Present

Built a web studio (byte.lunarflame.dev) from the ground up, bringing elegant web experiences to clients

- Delivering static and full-stack client sites in Svelte and React, with Figma-based designs
- Managing client engagement with constant feedback, a phased delivery system and change-order handling
- Drafting comprehensive client agreements, invoices, and recurring hosting subscriptions

Peer Learning Assistant - Worcester Polytechnic Institute | Worcester, MA

May 2026 - Present

A contact point for students and assistant for primary instructors regarding a wide range of course facilities

- Host frequent office hours to offer guidance on coursework while challenging the student's understanding
- Offer asynchronous help online and grade submitted coursework with comprehensive feedback

Undergraduate Researcher - WPI MPR Lab | Worcester, MA

October 2024 - December 2025

Experimented with the effectiveness of voice coil actuators in robotic percussion modules

- Led software, developing controls in C++ that intuitively modeled physical robotic percussion systems
- Contributed to project planning and development under the guidance of Professor Scott D. Barton via weekly meetings, video demonstrations, and a detailed lab notebook (youtu.be/tTjyxL7lbP4)

SKILLS

Languages: C/C++, Zig, Lua, Rust, Java, TypeScript/JavaScript, HTML/CSS, Bash, Python, Haskell, SQL

Tools: GNU/Linux, Wayland/X11, Git, Bun/Node, Svelte, React/Next.JS, Typst, AWS, MongoDB, Figma

Relevant Coursework: Computer Networks, Machine Organization, Operating Systems, Programming Language Design, Algorithms: Analysis and Design, Software Engineering